Midterm Exam (part 1) - Computational Physics I

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Date: Thursday 3 April 2025 **Duration:** 45 minutes

Credits: 8 points (4 questions) **Type of evaluation:** MT

Please provide concise answers to the following items:

1. **(2 points) Python Input/Output**
   a. Describe 2 distinct methods suitable for handling tabulated data input and output in Python.
   b. Explain one advantage and one disadvantage of each method.

We can use the native python function open("file_path", "r") or we could use the third library pandas as pandas.read_csv("file_path", sep="separation").
Pandas enables us to open and save tabulated data in a straightforward way, as well as getting the stored arrays. However, if we're not working in Python we could not use it.
The advantage of learning the native python method is that it is very similar in other programming languages. One disadvantage is that it is not as straightforward to carry out.

2. **(2 points) Data fitting methods**
   a. List the main steps to carry out a meaningful regression using empirical data in Python.
   b. Explain the difference between the least-squares (LM) and the χ^2 data fitting methods.

(1) Inspect the data. Do a first plotting attempt and look for a relation (monotonicity, linearity).

(2) Propose a physically motivated model to describe the data. It should have free parameters.

(3) Carry out an optimization routine to minimize a functional e.g. distances^2 ~~the discrepancies~~ between the model and the empirical data, thus restricting the free parameters.

(4) Compare your fitted model and the empirical data.

The main difference is that the χ^2 method does take into account the uncertainty associated with each data point. Both rely on mimizing an hypersurface but its form is different.

L.M: $\sum_{i=1}^{N}(y_i - f_{model})^2$ | χ^2: $\sum_{i=1}^{N}\frac{(y_i - f_{model})^2}{\sigma^2}$

3. **(2 points) Numerical differentiation**

a. Explain how the finite-difference methods for calculating derivatives work.

b. Provide the mathematical definition of a central-difference scheme.

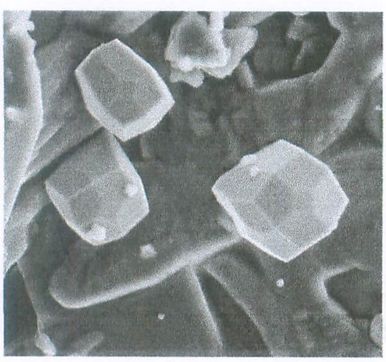
They rely on discretizing the definition of the derivative by doing a Taylor expansion of the function and computing the differences between neighboring data points, according to a step size.

The central difference method is of second order (according to the truncation error in the series). It's defined as,

$$\frac{df}{dx} \approx \frac{f_{i+1} - f_{i-1}}{x_{i+1} - x_{i-1}} + \mathcal{O}(\Delta x^2) \quad , \Delta x = x_{i+1} - x_{i-1}$$

4. **(2 points) Image processing**

-0.5



Imagine you obtain this photograph of iron crystals from a scanning electron microscope (credits: NASA/JSC), and you are asked to isolate the more prominent crystals from the background and from the rest of the image. Design and sketch a suitable algorithm workflow to achieve this goal in Python.

Isolate prominent iron crystals

I/O: Read the image as an array. (Select just one layer)

Do a first plotting attempt to check the intensity values of the crystals pixels. (# .imshow( ))

Flatten the array and make a 1D histogram to select an appropiate threshold value for the intensity

Thresholding: (Binary) If pixel intensity > threshold, assign a value of 1; otherwise zero.

np.where( )

multiple attempts to select the right threshold value

→ Make this a thresholding function.

Is there only one region left?

No ?

Sum all the pixels.

Select the region with more pixels and create a subset ?

Why subsetting?

Not sure why this is needed

Yes

Remove the subset from the array and store it how?

We have isolated the prominent crystals.

Return cleaned image.